Journal of Physics: Conference Series

The history and trends of semiconductor materials' development

To cite this article: Jitao Song 2023 *J. Phys.: Conf. Ser.* **2608** 012019

The history and trends of semiconductor materials' development

Jitao Song

University of Electronic Science and Technology of China, Chengdu, 610000

sjt13593315895@163.com

Abstract: Energy, materials and information are considered to be the three pillars of the emerging technological revolution. In terms of materials, the progress of electronic materials is particularly remarkable. A giant leap forward has been made in the development of all aspects of modern science and technology by the advent of computer, which features large scale and very large scale integrated circuits. A sequence of epoch-making semiconductor materials are constantly emerging, and rapidly become productivity and production tools, which has greatly promoted the rapid development of integrated circuit industry. The development level of semiconductor integrated circuits is one of the main indicators to measure a country's economic strength and scientific and technological progress. However, semiconductor materials are an important cornerstone of the development of integrated circuits. With the increasing number of electronic devices in all walks of life. the value of semiconductor materials is widely recognized. This paper introduces the characteristics, classification and application of semiconductor material, briefly summarizes the history of semiconductor materials from 1950s to now, and combined with its development process, studies the key factors to promote the development of semiconductor materials: the development of semiconductor materials is performance-oriented and limited by production costs. This paper puts forward the problems faced by today's semiconductor materials, and analyzes the current trend of diversified development of semiconductor materials. This paper can help readers quickly understand the development of semiconductor materials and the important factors affecting their development, and seek more solutions for the construction and development of the semiconductor industry.

Keywords: Semiconductor Materials, History and Development, Silicon, Germanium, Gallium Nitride, Silicon Carbide.

1. Introduction

Semiconductor material is a key material in modern electronic products and an important part of many commonly used electronic devices, including smart phones, tablet computers, etc. Moore's law has accelerated human development, but it will also hit a bottleneck. that chip manufacturers have used various means to keep pace with Moore's law, but it is still impossible to avoid the fact that the doubling effect of Moore's law has begun to slow down. There will always be a physical limit to constantly reducing the size of chips. Therefore, it is necessary to explore the trends of semiconductor materials development and seek new solutions for the continuation of Moore's law. Based on the development of semiconductor materials, this paper makes every effort to explore the crucial the key factors to promote the development of semiconductor materials. We should explore the development trend of semiconductor materials, clarify its development prospects and trends, and provide important references for future industry construction.

2. Introduction of semiconductor materials

2.1. Characteristics of semiconductor materials

The matter in nature can be divided into three categories of materials: conductors, semiconductors and insulators according to their electrical conductivity. The conductivity of insulators is very low, and its conductivity range is about between $10^{-18}\mathrm{S/cm} \sim 10^{-8}\mathrm{S/cm}$, such as fused quartz and glass. The conductivity of the conductor is high, and its conductivity range is about between $10^{4}\mathrm{S/cm} \sim 10^{6}\mathrm{S/cm}$, such as aluminum silver and other metals. However, the conductivity range of semiconductor material is between insulator and conductor. The conductivity of semiconductor materials is easily affected by temperature, trace impurity atoms, light and magnetic field. It is the high sensitivity of semiconductors to conductivity that determines the physical basis for making various semiconductor devices, making semiconductors one of the most important materials in various electronic applications.

2.2. Types of semiconductor materials

According to their chemical composition, structure and performance, semiconductor materials can be divided into 5 categories, including: element semiconductor, inorganic compound conductor, organic compound semiconductor, amorphous semiconductor and liquid semiconductor [1]. Silicon and germanium are elemental semiconductors composed of a single atom, which are group IV elements of the periodic table. Semiconductors composed of two or more elements are called compound semiconductors, such as silicon carbide (SIC), gallium arsenide (GaAs), etc.

Table 1 shows most of the known semiconductor materials which are also classified into various categories. It is easy to see from the table that semiconductor materials are basically composed of 12 elements with semiconductor properties (B, C, Si, P, S, Ge, As, Se, Sn, Sb, Te, I) in the periodic table. Some of them are unstable and volatile, namely: S, P, As, Sb and I, Gray Sn is easily transformed into white Sn at room temperature and becomes a metal; the melting point of B and C is too high to be processed into single crystals; Te is very scarce. In this way, only Se, Ge and Si are available for human industry in elemental semiconductors. Compound semiconductors are constituted by some compounds, which are from group I and groups V, VI, VII; Group II and groups IV, V, VI, VII; III and V, VI; IV and IV, VI; V and VI; Group VI and group VI. However, few of them have practical value or reached the practical stage in technology, industry semiconductor materials are mainly concentrated in III-V and II-VI compounds and their multiple solid solutions [2].

Table 1. Types of semiconductor materials.

Classification	**Semiconductor**	
	Symbol	**Name**
Element semiconductor	Si	silicon
	Ge	germanium
	Se	Selenium
Binary compound semiconductor	SiC	silicon carbide
	AlP	Aluminium phosphide
	AlAs	Aluminium arsenide
	AlSb	Aluminium antimonide
	GaN	Gallium nitride
	GaP	Gallium phosphide
	GaAs	Gallium arsenide
	GaSb	Gallium antimonide
	InP	Indium phosphide
	InAs	Indium arsenide
	InSb	Indium antimonide
	ZnO	Zinc oxide
	ZnS	Zinc sulfide
	ZnSe	Zinc stannate
	ZnTe	Zinc telluride
	CdS	Cadmium sulfide
	CdSe	Cadmium stannate
	CdTe	Cadmium telluride
	HgS	Mercury sulfide
	PbS	Lead sulfide
	PbSe	Lead stannate
	PbTe	Lead telluride
Ternary compound semiconductor	$AlGa_{11}As$	Gallium aluminum arsenide
	$AlIn_{11}As$	Indium aluminum arsenide
	$GaAs_{1-x}P_x$	Arsenic gallium phosphide
	$Ga_xIn_{1-x}As$	Indium gallium arsenide
	$Ga_xIn_{1-x}P$	Indium gallium phosphide
Quaternary compound semiconductor	$Al_xGa_{1-x}As_ySb_{1-y}$	Arsenic gallium aluminum antimonide
	$Ga_xIn_{1-x}As_{1-y}P_y$	Arsenic indium gallium phosphide

2.3. Application of semiconductor materials

Early application of semiconductor materials: the first application of semiconductors is the detector, a point-contact diode, which takes advantage of the rectifying effect of the semiconductor material. In addition, semiconductors were also used as rectifiers, photovoltaic cells, infrared detectors, etc. From 1907 to 1927, American physicists developed crystal rectifiers, selenium rectifiers and cuprous oxide rectifiers. In 1931, Lange and Bergman successfully developed selenium photovoltaic cells. In 1932, Germany successively developed semiconductor infrared detectors such as lead sulfide, lead selenide and lead telluride, which were used to detect aircraft and ships in World War II. During World War II, the Allied forces also made great achievements in the research of semiconductors. Britain used infrared detectors to detect German aircraft for many times.

Today, semiconductors have been widely used in household appliances, communications, industrial manufacturing, aviation, aerospace and other fields. Some components, made of semiconductor materials, provide a huge support for large-scale integrated circuits，meanwhile facilitating the runing speed or stability of equipment. Under normal circumstances, the conductivity of semiconductor electronic materials is fixed, and the conductivity performance is also good. However, if the temperature of the material continues to increase, the conductivity will increase accordingly. This characteristic provides the basic conditions for the realization of special application functions. Part of the thermistors adopts such characteristics of semiconductor materials, realizing the performance which can change with temperature [3]. At the same time, some impurities can be added to the semiconductor to form a PN junction, which provides basic support for the production of two pole and three pole components. The electrical properties of some semiconductor materials change according to light conditions, so they can be used to make photoresistors and realize special functions. In addition, there are some semiconductor materials that can realize the temperature difference effect and can be used to make special materials such as refrigerants. It can be considered that semiconductor materials are widely used and have great potential in the aspect of social level [4].

3. History of semiconductor materials

3.1. Germanium defeated silicon in the 1950s

The first transistor was invented by experts such as William Shockley, John Bardeen and Walter housebrattain in 1947. In 1958, Jack Kilby installed resistors, capacitors and transistors on a chip and invented the first integrated circuit [5]. They all use germanium in the end. One important reason is that the electron mobility of germanium is much higher than that of silicon. So the computing speed of computers made of germanium transistors is much faster than that of computers made of silicon transistors.

But germanium is not perfect. When the temperature exceeds 75℃, its performance will be greatly reduced. In contrast, the performance of silicon can remain unchanged at 170 degrees. The root cause of this difference is band gap, which is usually used as a important criterion to judge the performance of semiconductor (electrons move around the nucleus and form multiple orbits from the inside to the outside. The more outside the orbits, the higher the energy of the electrons. Covalent bonds can be formed between the outer electrons of different atoms. These electrons are called valence electrons, and their energy range is also called the valence band. When these valence electrons obtain enough energy from the outside, they can get rid of the nuclear bond and become free electrons. The energy range of these free electrons is called the conduction band. The energy difference between conduction band and valence band is called the band gap. The wider the band gap, the greater the energy difference between the conduction band and the valence band. The more difficult it is for valence electrons to become free electrons, and the environment of high temperature and high pressure is essentially the energy of electrons from the outside, so the wider the band gap of a material, the higher the voltage and temperature it can withstand.)

The band gap of silicon is much larger than that of germanium, so silicon is more heat-resistant than germanium. In general, the electron mobility of germanium is higher, while the metabolism of silicon is wider. In terms of chemical properties, each has its own merits. But germanium won because it won the price war. Price can adjust the allocation of resources, thereby affecting supply and demand, and guide production and consumption. In the 1950s, the main reason why silicon was defeated by germanium, whose content in the earth's crust was far less than its own, was that it was not as cheap as germanium. In 1959, the unit price of silicon transistor was $14.53, while that of germanium transistor was only $1.96.

3.2. Unshakable status of silicon in the 1960s and 1990s

By the 1960s and 1970s, the manufacturing process of silicon had made a breakthrough, and the production cost had dropped. Due to the high content of silicon in the earth's crust and good performance, silicon became the cornerstone of the semiconductor industry. Although later semiconductors were used more and more, such as integrated circuits, communication systems, photovoltaic power generation, high-power power conversion and so on, silicon can meet these requirements. At that time, some new semiconductor materials (second-generation semiconductor materials) were also applied, mainly gallium arsenide, indium phosphide and other materials. These materials have excellent properties, such as good electron mobility, band gap and other characteristics. They are suitable for high-speed, high-frequency, high-temperature and high-power electronic devices. They are excellent materials for making high-performance microwave and millimeter wave devices and circuits, and are widely used in phased array radar, electronic countermeasures, satellite communications, mobile communications and other fields have broad dual-use market and development prospects [6]. However, this kind of material is not only scarce, but also has strong toxicity, which is easy to cause serious pollution to the ecological environment. So the status of silicon is still unshakable.

4. Difficulties and trends in the development of semiconductor materials

4.1. Problems faced by the development of semiconductor materials today

Nowadays, with the rapid development of scientific and technological innovation, modern Internet, communication, microelectronics and other emerging technologies, the performance requirements of semiconductor materials are becoming higher and higher, and silicon has been unable to meet the demand. Especially in the two major categories of semiconductors, power semiconductors and chip semiconductors, silicon has encountered a bottleneck that is difficult to overcome.

Power semiconductors are the core devices of current switches in circuits and power electronic conversion devices, and are generally used in communications, new energy, automobiles, and high-speed rail. It is generally used in communication, new energy, automobile and high-speed railway. Power devices are involved in every link of the power value chain, from power generation (e.g., photovoltaic power plants) to battery energy storage through uninterruptible power supplies to the final load point. For example, a large number of RF power semiconductor devices are used in the 5g base station. A few decades ago, silicon could meet the needs of power semiconductors. However, with the development of technology, the working environment of power semiconductors has become more and more harsh. Nowadays, in the application scenarios of high-speed rail and wind power, power semiconductors often need to withstand thousands of volts of high voltage, while silicon-based transistors cannot operate normally in this environment.

For chip semiconductors, in order to reduce power consumption and improve integrated circuit technology, the feature size of the world's most advanced integrated circuits has reached the level of three nanometers. If the size is further reduced, silicon-based semiconductors will have problems such as carrier surface scattering, velocity saturation, ionization and hot electron effect, which are collectively referred to as short channel effect. In short, because the channel is too short, the mutual interference of electric fields in multiple directions in the channel is intensified, so that the gate is not tightly closed, and the transistor will have leakage, heating and other problems. This part of energy consumption, sometimes even accounts for half of the total energy consumption of the chip.

Power semiconductors need wider band gaps to withstand more extreme conditions. However, for chip semiconductors, a wider band gap is obvious fault.Because if the band gap becomes wider, the working current and voltage of the chip will become larger, which will cause the chip to heat up and power consumption, which is contrary to the requirement of reducing the characteristic size of the chip.

4.2. Development trend of current semiconductor materials

Since silicon cannot meet the application requirements of different scenarios, different characteristics of different materials are used to make up for the deficiency of silicon. In order to meet the harsh environment of power semiconductor devices, the third generation of semiconductor materials came into being. These materials are relatively stable in operation at high temperature, consume relatively small amount of power, and have high operation efficiency. They can operate stably even at high voltage and high frequency [7]. SiC and GaN based power semiconductor devices are regarded as the next generation of power semiconductor devices. Based on the commercial production of SiC rectifiers, various SiC power switching devices (such as MOSFET, MESFET, JFET, IGBT, SIT, GTO, BJT, etc.) have been successfully developed. Although SiC power switching devices have made great progress in recent years and the characteristics and power capacity of various SiC power switching devices have been continuously improved, the large-scale commercialization of SiC power switching devices still depends on the development of high-quality, large-area defect free and low-cost SiC materials, as well as the optimization of device structure and the study of device reliability mechanism. With the rapid development of LED applications, GaN materials and devices are becoming the development focus of power semiconductors. The device based on AlGaN/GaN material has lower on-resistance than SiC and the maturity and gradual commercialization of large-diameter GaN epitaxy technology, makes GaN power semiconductor develop from microwave power device to power rectifier, power switch and power integration [8]. In order to solve the short channel effect of chip semiconductors, two-dimensional materials and new channel semiconductor materials have been developed. Two dimensional materials are very conducive to the construction of high-performance ultra short channel field-effect transistors because of their physical limit of atomic layer thickness and the absence of hanging bonds on the surface. It provides the possibility to solve the important problem of transistor miniaturization bottleneck, and has great application potential in the future VLSI. The boron nitride / graphene heterostructure with undercut structure was fabricated by selective etching of boundary aligned boron nitride / graphene heterostructure with hydrogen plasma, and then the heterostructure was transferred to single-layer molybdenum disulfide to fabricate short channel MoS2 FET. The MoS2 transistor manufactured by this new manufacturing technology has a current density of up to $433\mu A/\mu m$ at $9nm$ and $VDS = 1V$ and a driving current of up to $734\mu A/\mu m$ at $VDS = 2V$. At the same time, the device has good switching characteristics, its switching ratio reaches 3×10^7, and the subthreshold swing is$120mV/dec$. In addition, the device shows good short channel immunity, and its leakage induced barrier is reduced DIBL ~ $50mV/V$[9]. In the future, the application of semiconductor materials will be further deepened, and new semiconductor materials will be more and more.

Although these materials have better performance in different application scenarios, the current status of silicon is still irreplaceable, because these materials are faced with the same problem as silicon in the past - price (cost). The cost of silicon carbide devices of the same specification is about 3-5 times that of silicon-based devices, and gallium nitride will be more expensive. The main reason is similar to that of silicon decades ago: the manufacturing process is immature. However, with the successful case of silicon, this problem must also be solved over time.

At present, the research of silicon carbide and gallium nitride technology needs to start from two aspects of crystal growth and processing, and continuously promote the enhancement and extension of the function of semiconductor materials. Taking silicon carbide as an example, the main preparation methods include physical vapor transport method and seed sublimation method. Due to the strong hardness of silicon carbide itself, diamond abrasive based processing equipment is mainly used to cut and grind silicon carbide materials. By strictly controlling the external environment such as equipment, pressure and temperature, the quality of single crystals can be improved and the cost of single crystals can be effectively reduced [10].

5. Conclusion

To sum up, the development of semiconductor materials is performance-oriented, limited by production costs, and at present, the semiconductor industry is facing the slowdown of the multiplier effect of Moore's law. Traditional semiconductor materials are gradually not used in the field of modern electronic technology, while new materials are facing the problem of high production costs. The author believes that semiconductor materials with better performance are bound to replace the former leaders, and semiconductor industry practitioners should actively explore new materials with better performance. At the same time, more in-depth research on technology and process is needed in order to improve the level of processing technology and reduce production costs. According to the specific research and application demands, select the corresponding types of materials, give full play to the performance advantages of semiconductor materials, and strive to continue the doubling effect of Moore's theorem. With the rapid development of Electronic Science and other technologies, the demand for the application of semiconductor materials has also increased significantly. By analyzing the history and development trend of semiconductor materials, readers can quickly understand the development of semiconductor materials and important factors affecting their development, and seek more solutions for the construction and development of the semiconductor industry.

References

[1] Yaoguang Yang. Application of semiconductor materials in Electronic Science and technology [J]. Electronic components and information technology, 2020, 4 (10): 5-6+8. doi:10.19772/j.cnki.2096-4455.2020.10.003
[2] Dongjiang He. Classification and application of semiconductor materials [J]. Metallurgical standardization and quality, 1998 (02): 39-42.
[3] Gou Wei. Development trend of semiconductor materials in Electronic Science and technology [J] Computer products and circulation, 2019, 000 (002): 78-79.
[4] Boyang Jiang. Discussion on the development trend of semiconductor materials in Electronic Science and technology [J]. China new communications, 2022, 24 (02): 73-74.
[5] Kangnian Yan. A historical review of the debate on the invention of integrated circuits and the right to invention -- Commemorating the 40th anniversary of the invention of integrated circuits [J]. Dialectics of nature newsletter, 1999 (02): 60-68.
[6] Hewei Li. Development trend of second generation semiconductor materials, devices and circuits [J]. World products and technology, 2003 (11): 36-39.
[7] Danzengpingcuo. Basic research and analysis in the development of semiconductor technology [J]. Electronic components and information technology, 2018, (11): 32-34, 82.
[8] Bo Zhang, Power semiconductor device technology [C]//. Abstracts of 2012 academic annual meeting of China Vacuum Society [publisher unknown], 2012:49.
[9] Jinpeng Tian, Research on high performance MoS2 FET and logic device [D]. University of Chinese Academy of Sciences (Institute of physics, Chinese Academy of Sciences), 2022. Doi:10.27604/d.cnki.gwlys.2022.000043.
[10] Xu Wang. Application status and development trend of semiconductor materials [J]. Light source and lighting, 2022 (01): 67-69.